

Bitwise differential trail through **SERPENT** found by SAT

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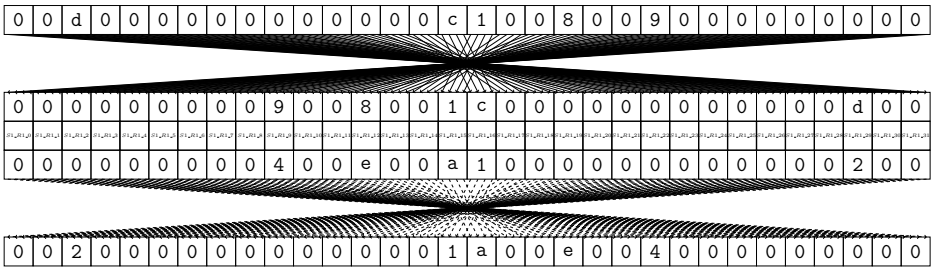
1 SERPENT

Maximal differential probability: 2^{-29}

0	0	d	0	0	0	0	0	0	0	0	0	0	0	0	c	1	0	0	8	0	0	9	0	0	0	0	0	0	0	0			
0	0	d	0	0	0	0	0	0	0	0	0	0	0	0	c	1	0	0	8	0	0	9	0	0	0	0	0	0	0	0			
R1																																	
0	0	d	0	0	0	0	0	0	0	0	0	0	0	0	c	1	0	0	8	0	0	9	0	0	0	0	0	0	0	0			
0	0	d	0	0	0	0	0	0	0	0	0	0	0	0	c	1	0	0	8	0	0	9	0	0	0	0	0	0	0	0			
R2																																	
0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	1	a	0	0	e	0	0	4	0	0	0	0	0	0	0	0			
0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	1	a	0	0	e	0	0	4	0	0	0	0	0	0	0	0			
R3																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	5	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	5	0		
R4																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	5	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	5	0		
R5																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	a	0	0	4	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	a	0	0	4	0		
R6																																	
0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
R7																																	
0	0	0	0	a	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0	0	0	0	a	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
R8																																	
0	0	0	2	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	8	1	0
0	0	0	2	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	8	1	0
R9																																	
0	0	0	2	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	8	1	0
0	0	0	2	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	8	1	0
R10																																	
0	0	0	6	0	0	0	b	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	0	0	0	c	7	0
0	0	0	6	0	0	0	b	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	0	0	0	c	7	0
R11																																	
8	9	7	4	4	5	3	8	0	1	2	4	1	4	9	0	a	5	1	c	b	0	8	6	5	8	3	1	1	3	4	6		
8	9	7	4	4	5	3	8	0	1	2	4	1	4	9	0	a	5	1	c	b	0	8	6	5	8	3	1	1	3	4	6		

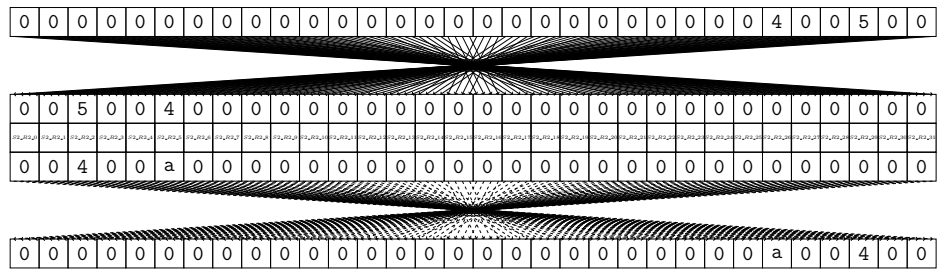
2 SBoxLayer_R1

Maximal differential probability: $2^{-11.0}$



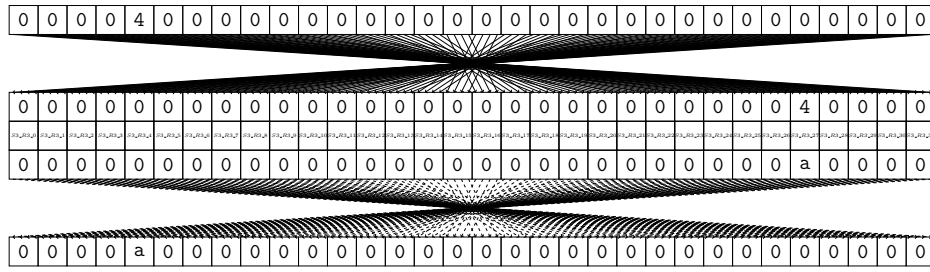
3 SBoxLayer_R2

Maximal differential probability: $2^{-5.0}$



4 SBoxLayer_R3

Maximal differential probability: $2^{-3.0}$



5 SBoxLayer_R4

Maximal differential probability: $2^{-10.0}$

